

WHAT BENDS THE CURVE?

Paris at ten: long trends, a new acceleration, and the national transformations that carry the signal

Structural breaks · Kaya pathways · five decades of national decarbonization

204 jurisdictions 97 eligible national breaks 7 mechanism families 9 constructive decarbonizations

THE QUESTION

Paris is working — but through what?

The good news is real.

Worst-case emissions scenarios have moved off the table, and global CO2 growth has slowed from +2.1% per year (1990–2015) to +0.4% per year since Paris.

The open question:

Is the global improvement a direct effect of the agreement — or the continuation of trends rooted decades earlier, which Paris coordinates and reinforces?

Our approach:

Rather than assuming treaty years are turning points, we estimate structural breaks from the data — globally and for every country with adequate annual series — then use historical evidence to ask why the trend changed.

Why intensity comes first

$$\begin{aligned} \text{growth of CO}_2 &= \text{growth of GDP} \\ &+ \text{change in C/E (fuel mix)} \\ &+ \text{change in E/GDP (efficiency)} \end{aligned}$$

Emissions cannot bend until an intensity slope accelerates. Intensity is the leading indicator: any policy signal must appear there first — on carbon or energy intensity, not necessarily on emissions.

Unless the intensity terms outpace GDP growth, emissions cannot stabilize.

What a breakpoint means — and the Kaya lens

The breakpoint model

$$y(t) = a + b \cdot t + d \cdot \max(t - k, 0) + e(t)$$

b is the pre-break slope; $b + d$ the post-break slope; d the change. The unrestricted model searches all admissible years for the strongest slope change; a moving-block bootstrap describes how stable that date is under serially dependent noise.

A breakpoint identifies when a trend changed — with uncertainty. It is not a peak year, not a policy-adoption date, and never a causal estimate by itself. Historical and external evidence answers why.

Inference: Newey–West standard errors for fixed-date tests; BIC for model evidence; moving-block bootstrap for date stability.

The Kaya decomposition

$$C/GDP = (C/E) \times (E/GDP)$$

C/GDP — aggregate carbon intensity of output: the observed outcome.

C/E — supply side: fuel mix, electricity systems, conversion technology.

E/GDP — demand side: efficiency, sectoral structure, economic composition.

We estimate all three, for the world aggregate and every country, with one harmonized procedure. The component pattern constrains which mechanisms are plausible.

From 204 jurisdictions to 97 eligible national breaks

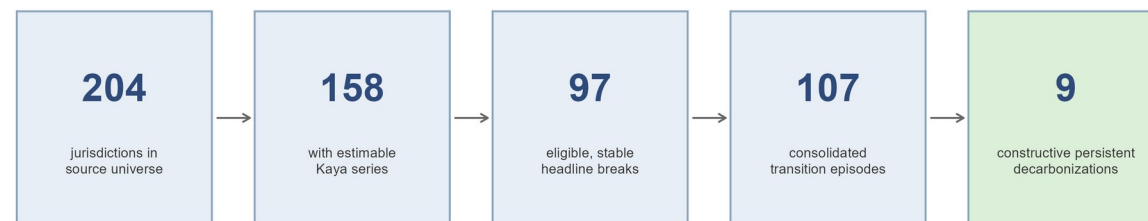
Eligibility is strict by design

- at least 40 annual observations, 10+ years on each side of the break, break 10+ years from the boundary
- BIC improvement of at least 6 over the unbroken trend
- 70%+ of bootstrap dates within 5 years; interval no wider than 12 years
- standardized slope change of at least 0.25

204 jurisdictions → 158 estimable → 97 countries with an eligible, stable headline break; nearby component breaks consolidate into 107 transition episodes (15-year grouping; 10/20-year sensitivity).

Treaty attribution ladder (all five required):
Newey-West significance → top 10% vs nearby placebo dates → best date within 3 years of the treaty → compatible unrestricted break → plausible domestic mechanism.

From the full country universe to credible decarbonization successes



9 breaks are policy-enabled; 9 (a partly different set) combine a persistent favorable bend with falling absolute CO2

Four acts

I

Is the global bend Paris?

Global curves bent in the 1970s; since ~2013 they are bending faster. The acceleration is real — its attribution is not yet identified.

II

What bends national curves?

97 eligible breaks scattered across five decades and seven historically validated mechanism families.

III

Did the bends count?

Transition quality separates genuine decarbonization from recessions, disruption, offshoring, and growth that outruns efficiency.

IV

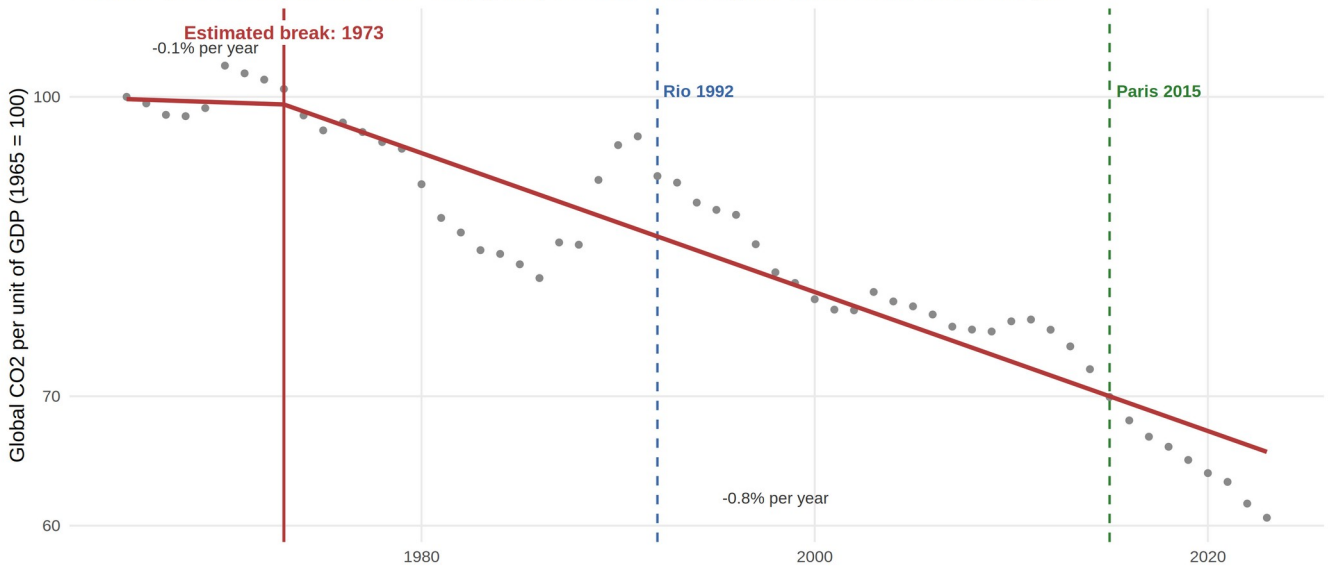
Can we trust the story?

Event-timing tests against chance, external ratings, consumption-based accounting — and unresolved cases reported as such.

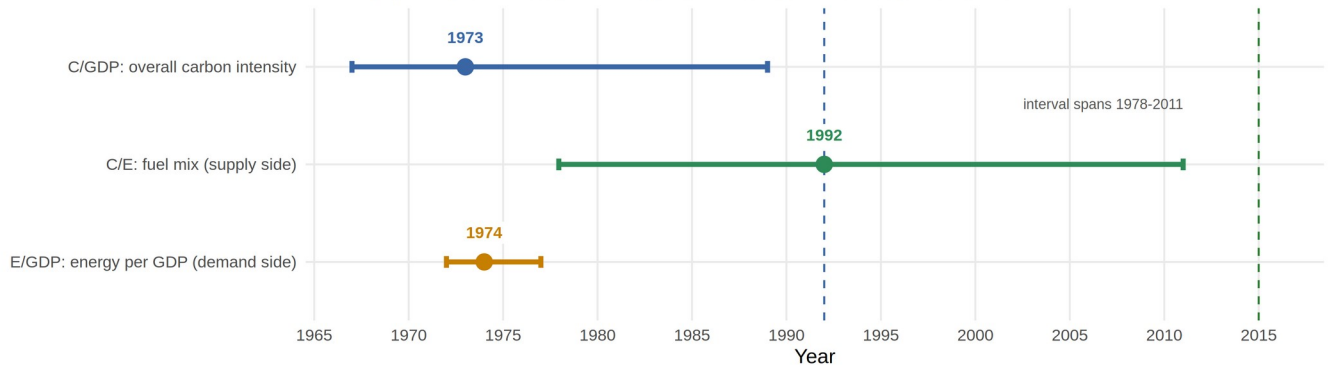
Global curves first bent in the 1970s — decades before the climate regime

The world's carbon-intensity curve bent in the early 1970s, decades before Rio or Paris

Global CO2 per unit of GDP, indexed to 1965 = 100 (log scale). The red line is a two-segment trend with a data-selected break year.



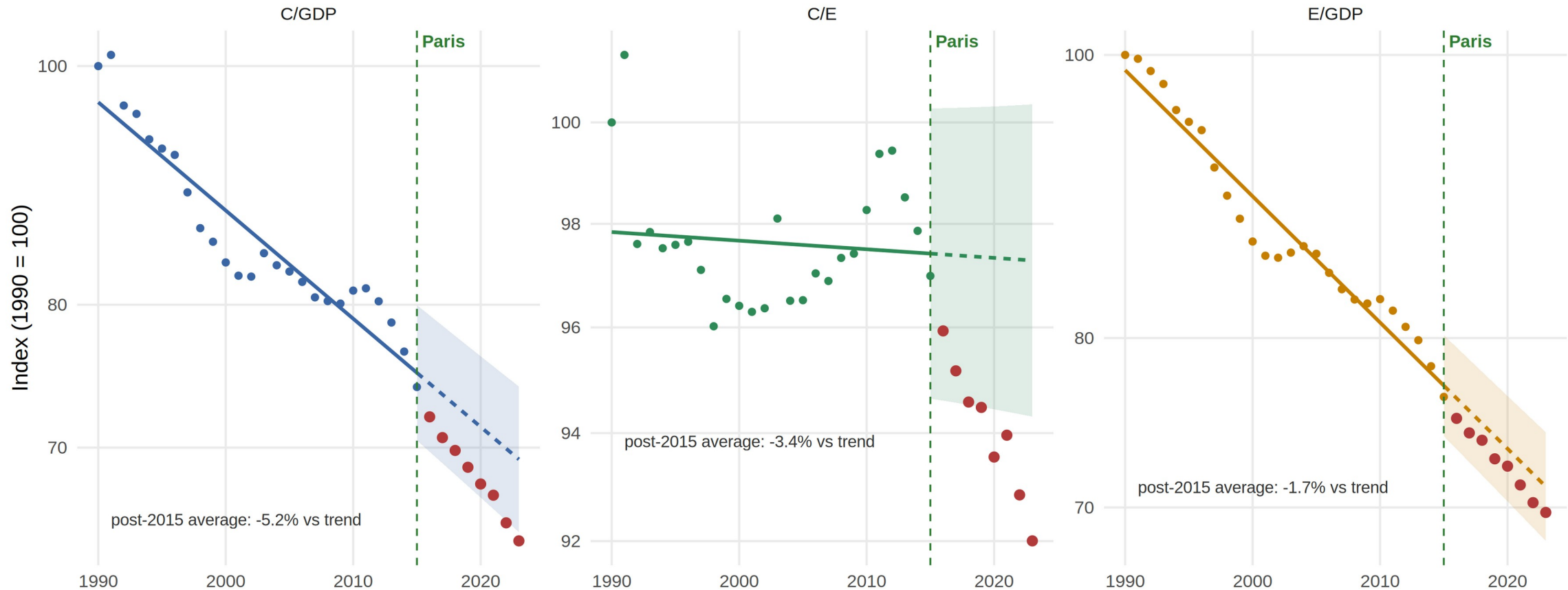
Each Kaya pathway's estimated global break year, with moving-block bootstrap 95% intervals



Paris at ten: intensity is outrunning its old trend — but the bend began earlier

Since Paris, global intensity is finally outrunning its old trend - but the bend began before the treaty

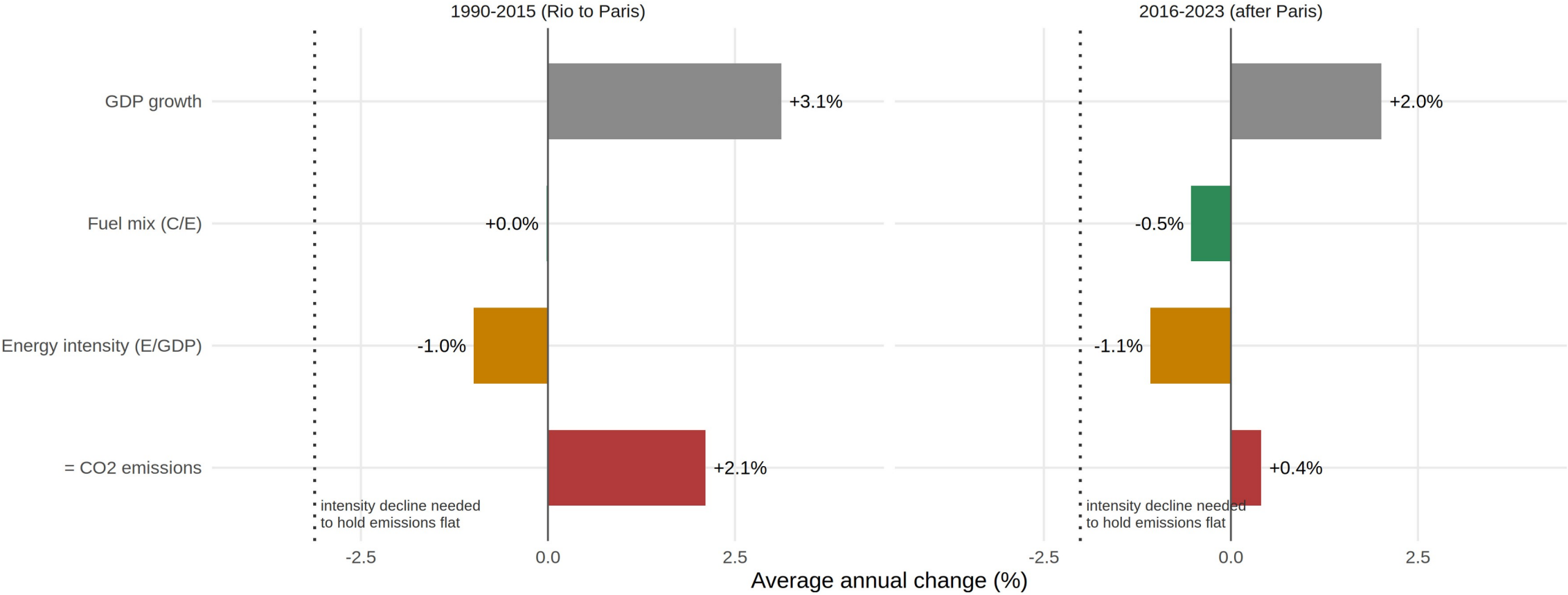
Each panel: 1990 = 100 (log scale). Solid line: 1990-2015 trend; dashed: its extrapolation with a 95% prediction band; red points: observed after Paris.
The fall below the band is a real supply-side acceleration (C/E -0.7 pp/yr, Newey-West $t = 4.3$; C/GDP -0.9, $t = 2.9$), but candidate onsets from 2011 to 2016 fit the full record about equally well (best: 2013), and the demand side (E/GDP -0.3, $t = 1.5$) shows no acceleration - so a Paris effect cannot yet be separated from the renewables cost decline that preceded it.



The arithmetic of bending the curve: intensity must fall as fast as GDP grows

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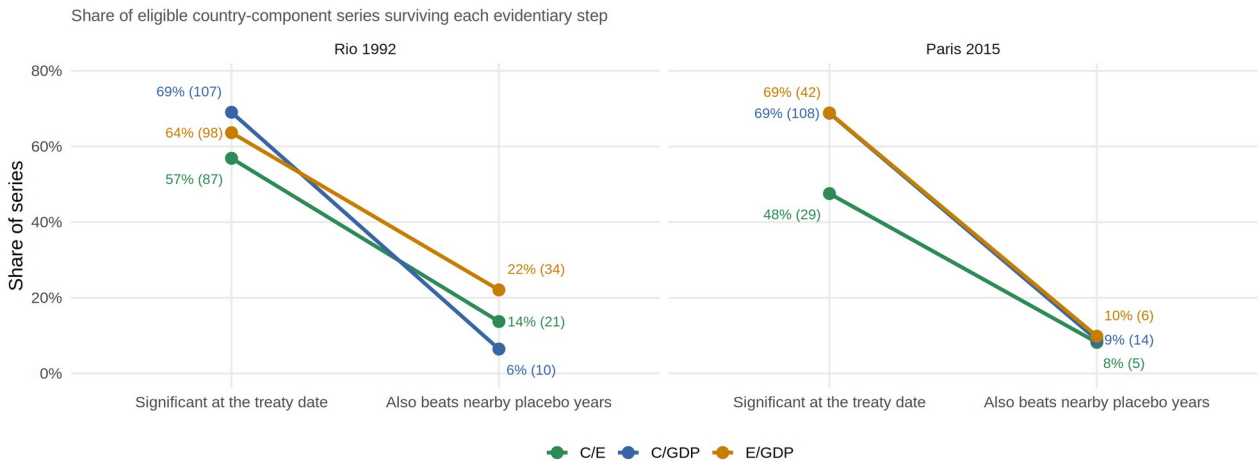
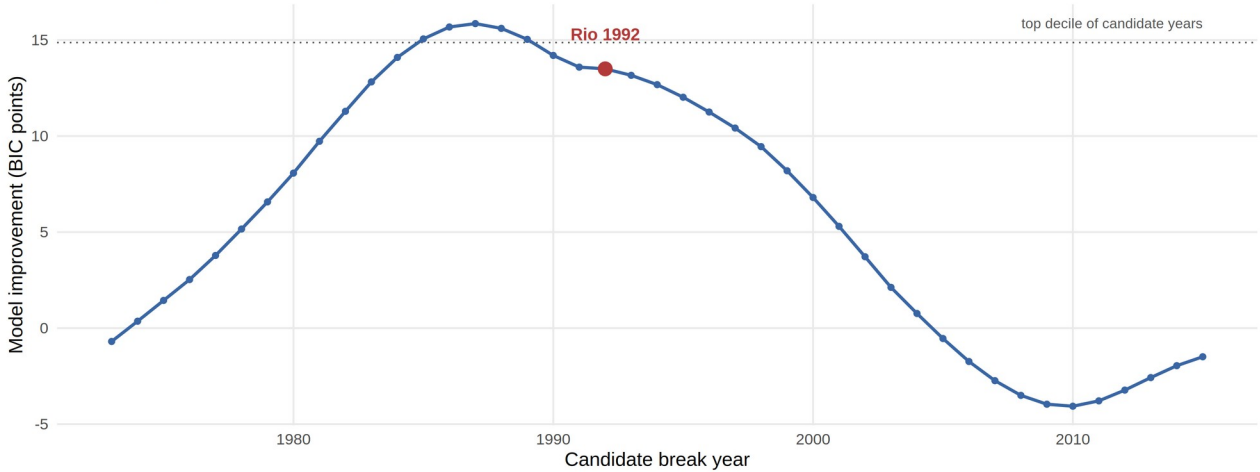
Average annual log growth rates of the global Kaya terms. CO2 growth = GDP growth + fuel-mix change + energy-intensity change, so the emissions curve cannot bend until the intensity slopes accelerate - which is where any policy signal must appear first. Emissions growth slowed from +2.1% to +0.4% per year after 2015: roughly two-thirds of that slowdown is slower GDP growth, one-third a cleaner fuel mix.



The placebo discipline: many years look like turning points

Why a significant treaty-year break is weak evidence on its own

BIC improvement from placing the global fuel-mix (C/E) break in each candidate year (aggregate reconstruction): many years fit comparably well, which is exactly why the bootstrap interval in Figure 1 is so wide



The global record can show an acceleration — not yet its author

What the data establishes

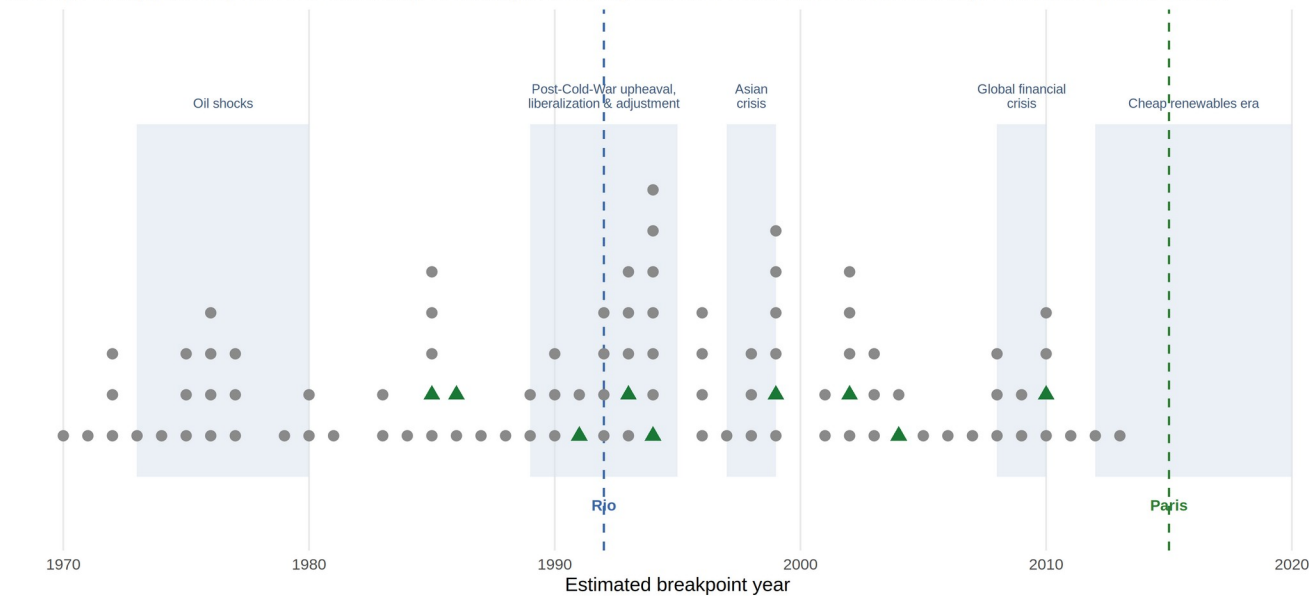
- A real post-2013 supply-side acceleration: C/E falling $-0.5\%/yr$ after a flat quarter-century; C/GDP -0.9 pp/yr faster than the Rio-to-Paris trend (Newey-West).
- Emissions growth slowed from $+2.1$ to $+0.4\%/yr$ — the direction an effective Paris requires.
- No acceleration on the demand side (E/GDP): efficiency and structure continue their 50-year trend.

Why attribution stays open

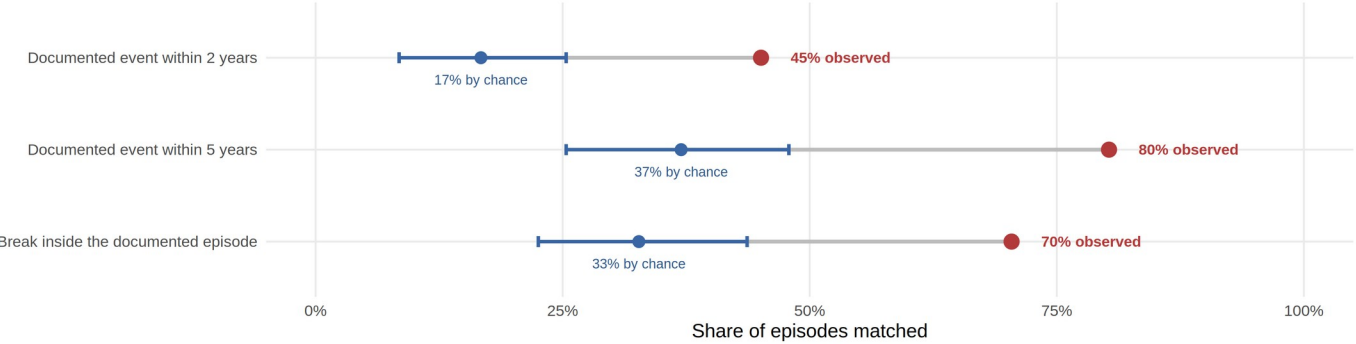
- Timing: candidate onsets 2011–2016 fit about equally well (best 2013) — indistinguishable from the renewables cost collapse that preceded Paris.
- Power: with 8 post-Paris years, only an acceleration ≈ 0.8 pp/yr is detectable at 80% power — Paris would have had to double a 50-year trend to prove itself already.
- Aggregation: China's WTO-era coal boom flattened the global fuel-mix baseline; the world series is a China-weighted average that hides national signals.
- Design: Paris works through NDCs and national transformation — its signature should appear country by country. So we look there.

97 national breaks across five decades — tracking economic history

97 national breakpoints across five decades: tracking economic history, not treaty dates
One dot per country (statistically selected headline break). Green triangles: the 9 policy-enabled low-carbon cases. Shaded bands: major economic and political episodes.



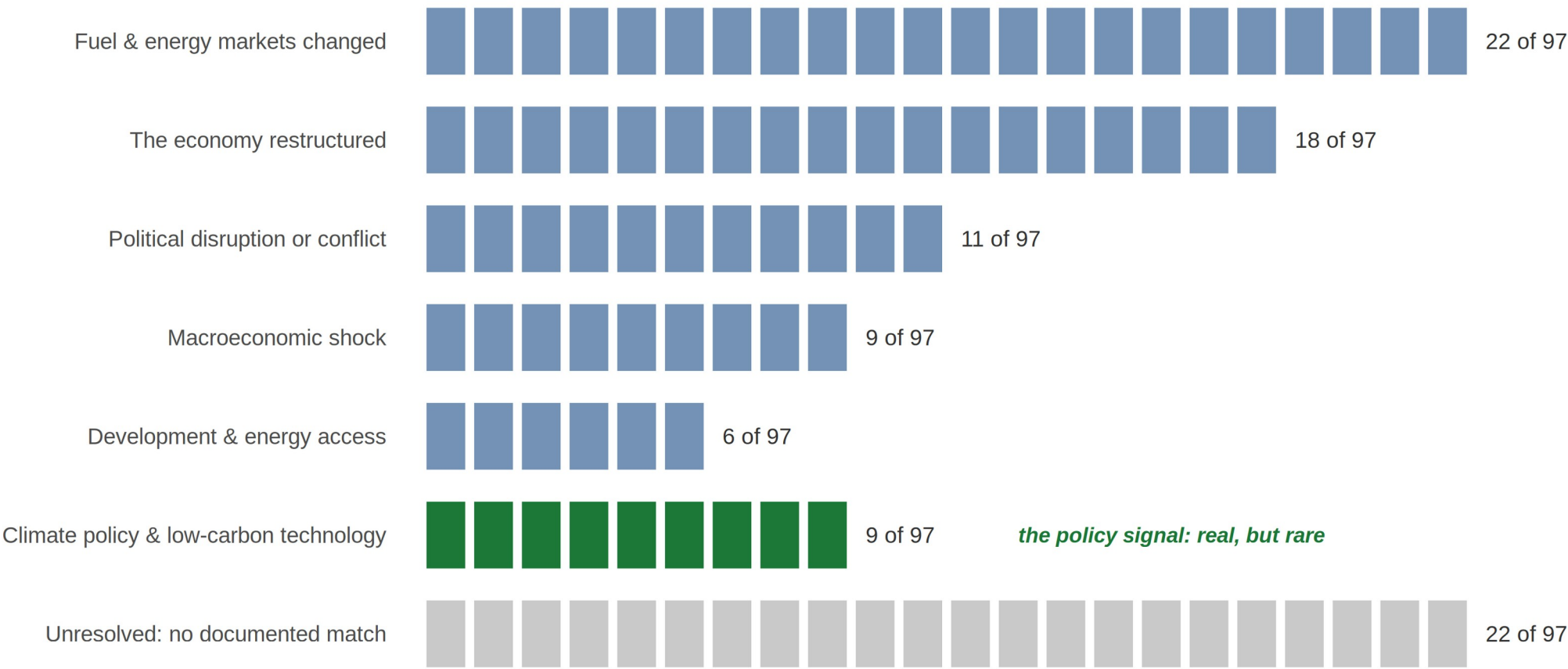
And the breaks are not arbitrary: among 71 documented episodes, historical events sit near the estimated breaks far more often than chance



What accompanied the bends: mostly markets, restructuring, disruption

What accompanied the 97 bends? Mostly markets, restructuring, disruption

One square per country: the best-supported historical mechanism for each statistically selected headline breakpoint



Seven mechanisms, six verdicts

MECHANISMS — what accompanied the bend

Policy-enabled low carbon — deliberate climate/energy policy made physical: infrastructure, fuel mix, market design.

Fuel & energy markets — market-driven fuel switching or supply transformation.

Economic restructuring — industry to services, liberalization, post-socialist transition.

Macroeconomic shock — recessions and financial crises that cut energy use.

Political disruption — wars, state collapse, sanctions, regime change.

Development & energy access — electrification and modernization.

No attributed mechanism — a real, eligible break the record cannot yet explain (22 of 97).

VERDICTS — four tests of the outcome

favorable? persistent? absolute CO2 falling? constructive mechanism?

Constructive persistent decarbonization (9) — passes all four — the gold standard.

Persistent fuel substitution (13) — durable supply-side switch; narrower transformation.

Intensity gain; CO2 still rising (12) — relative gain overwhelmed by growth.

Contractionary / disruptive (20) — the 'improvement' is a crisis in disguise.

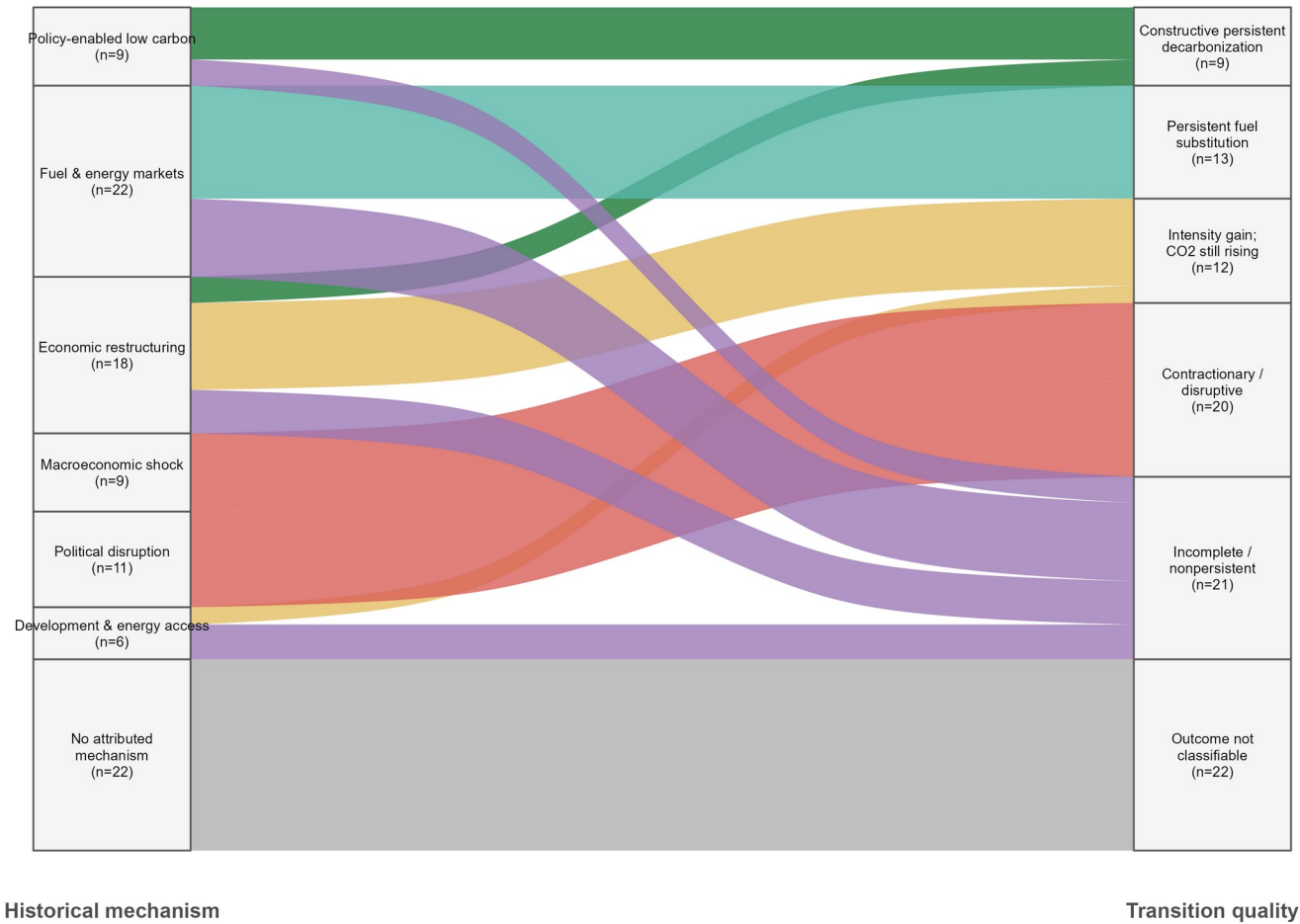
Incomplete / nonpersistent (21) — the bend reversed or did not hold.

Outcome not classifiable (22) — too little post-break evidence.

From mechanism to verdict: 97 countries flow to six outcomes

A favorable bend is not automatically a successful transition

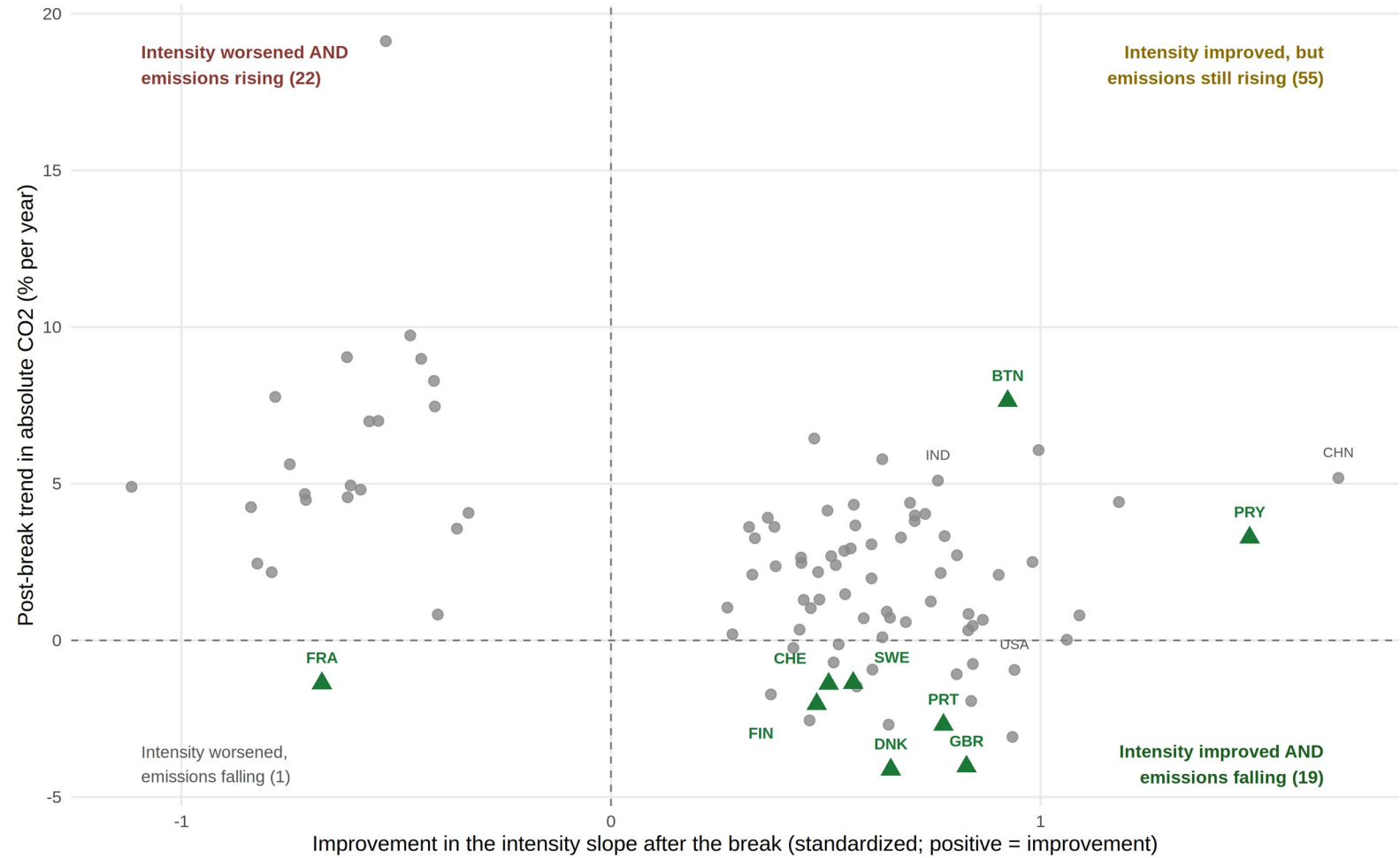
Flows of 97 countries from the mechanism accompanying the break to the realized transition-quality classification; residual categories shown at the bottom of each axis



Lower carbon intensity is not the same as falling emissions

A favorable bend is not yet a success: intensity often improves while emissions keep rising

Green triangles: the 9 policy-enabled cases. Only 9 of 97 countries qualify as constructive, persistent decarbonization once persistence and mechanism are also required.

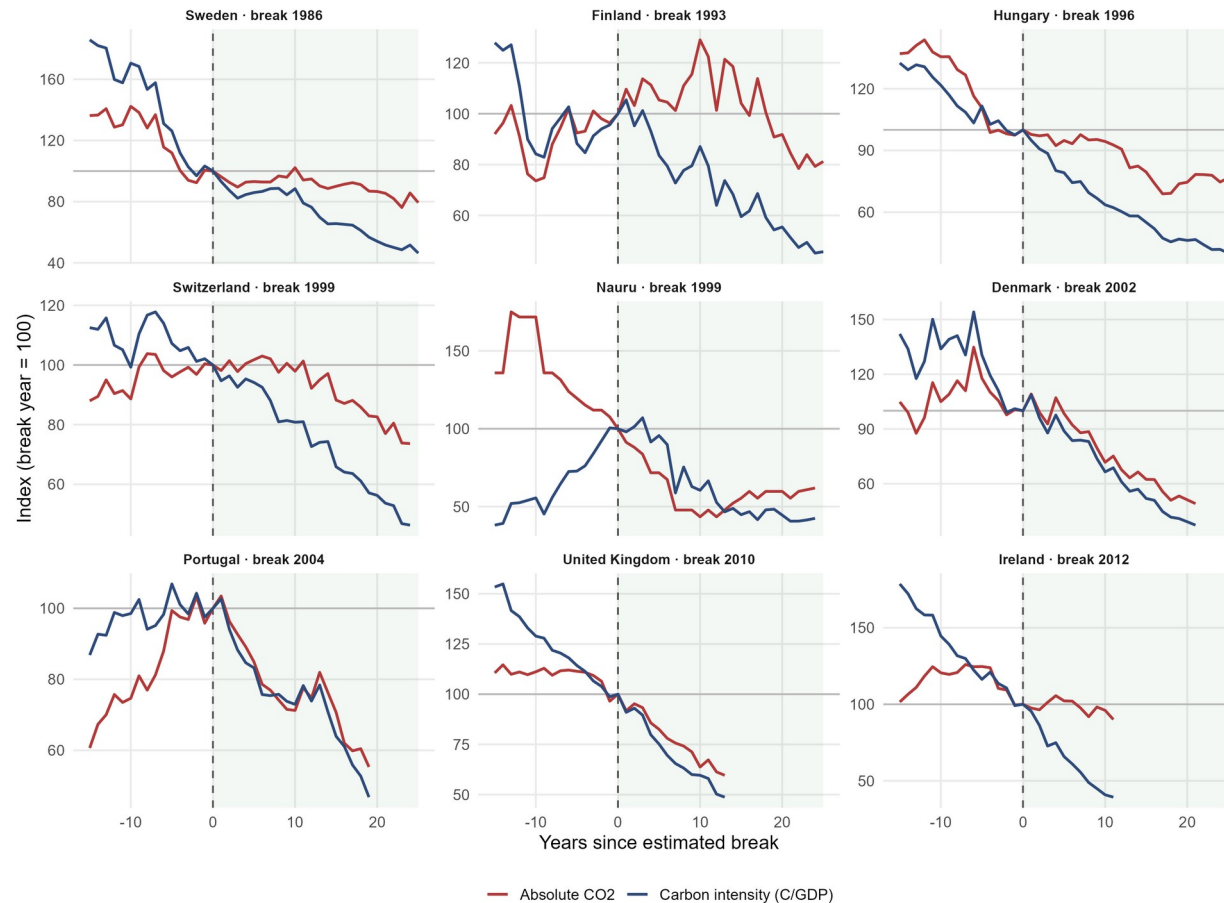


The nine constructive persistent decarbonizations

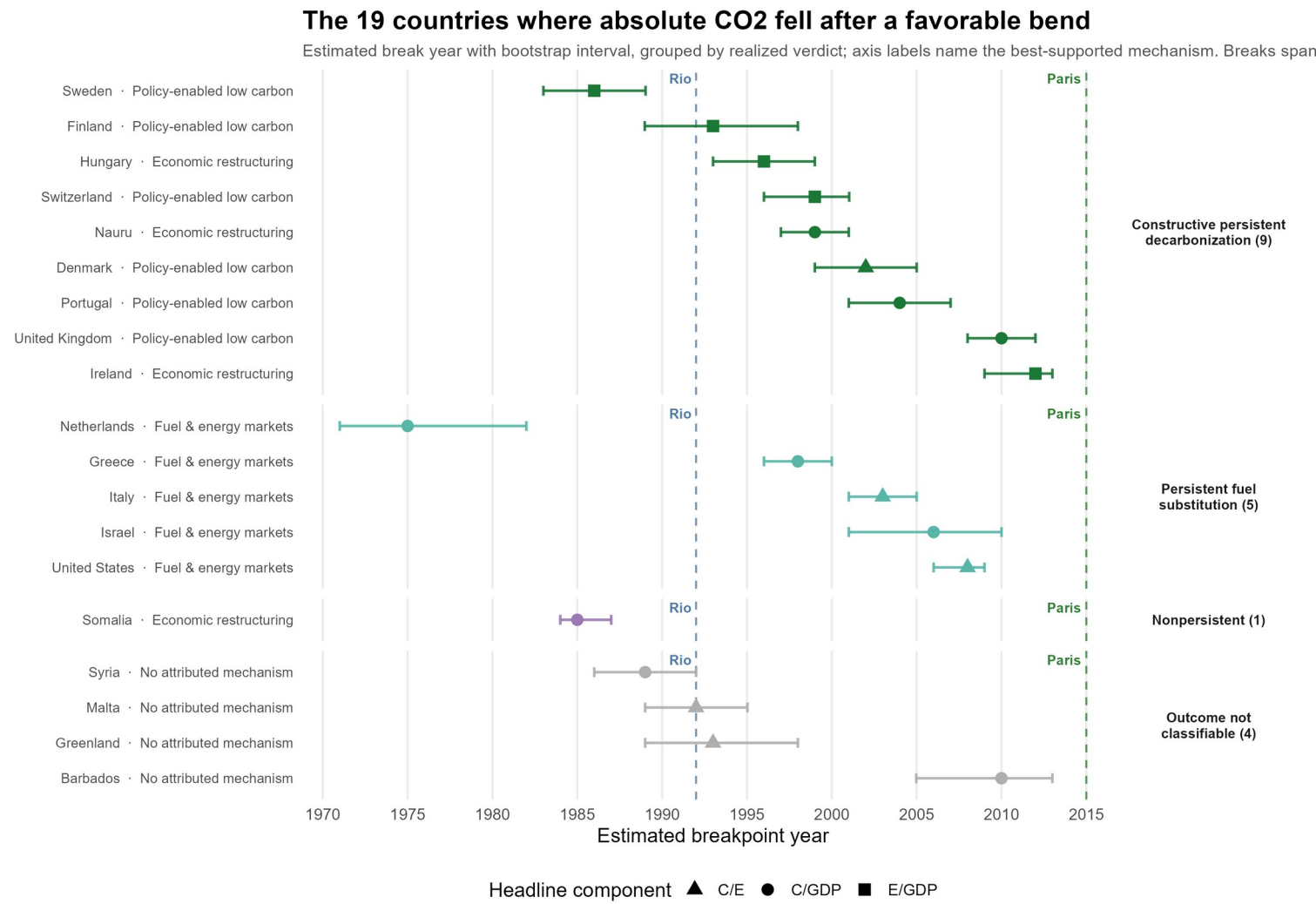
Sweden 1986 · Finland 1993 · Hungary 1996 · Switzerland 1999 · Nauru 1999 · Denmark 2002 · Portugal 2004 · United Kingdom 2010 · Ireland 2012

The nine constructive persistent decarbonizations

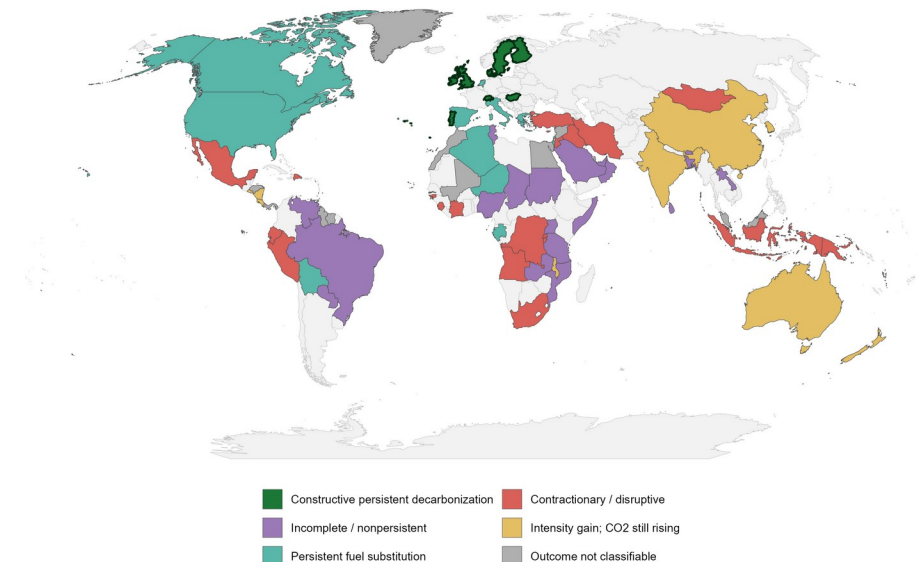
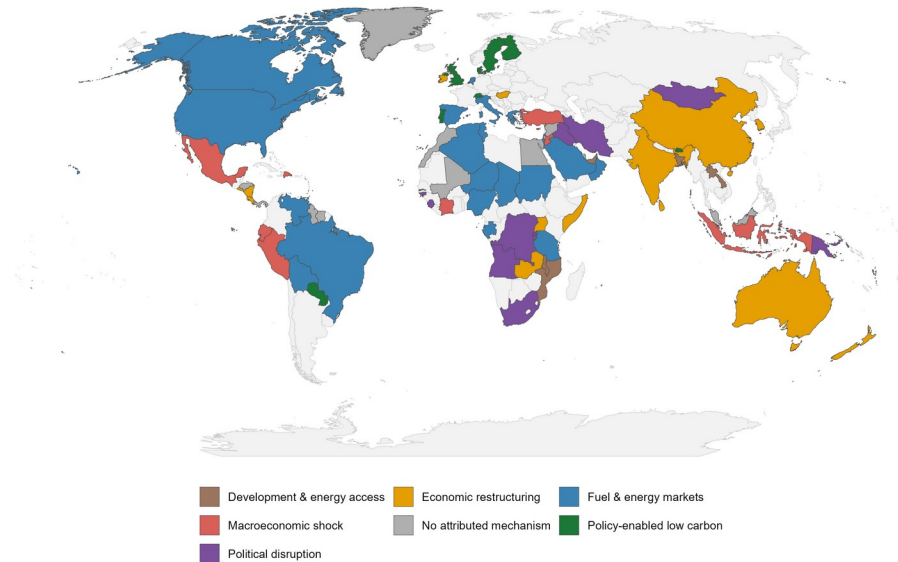
Absolute CO2 and carbon intensity indexed to 100 at each country's estimated break year; both decline after the break (shaded region)



Nineteen countries cut emissions — ten don't earn the verdict



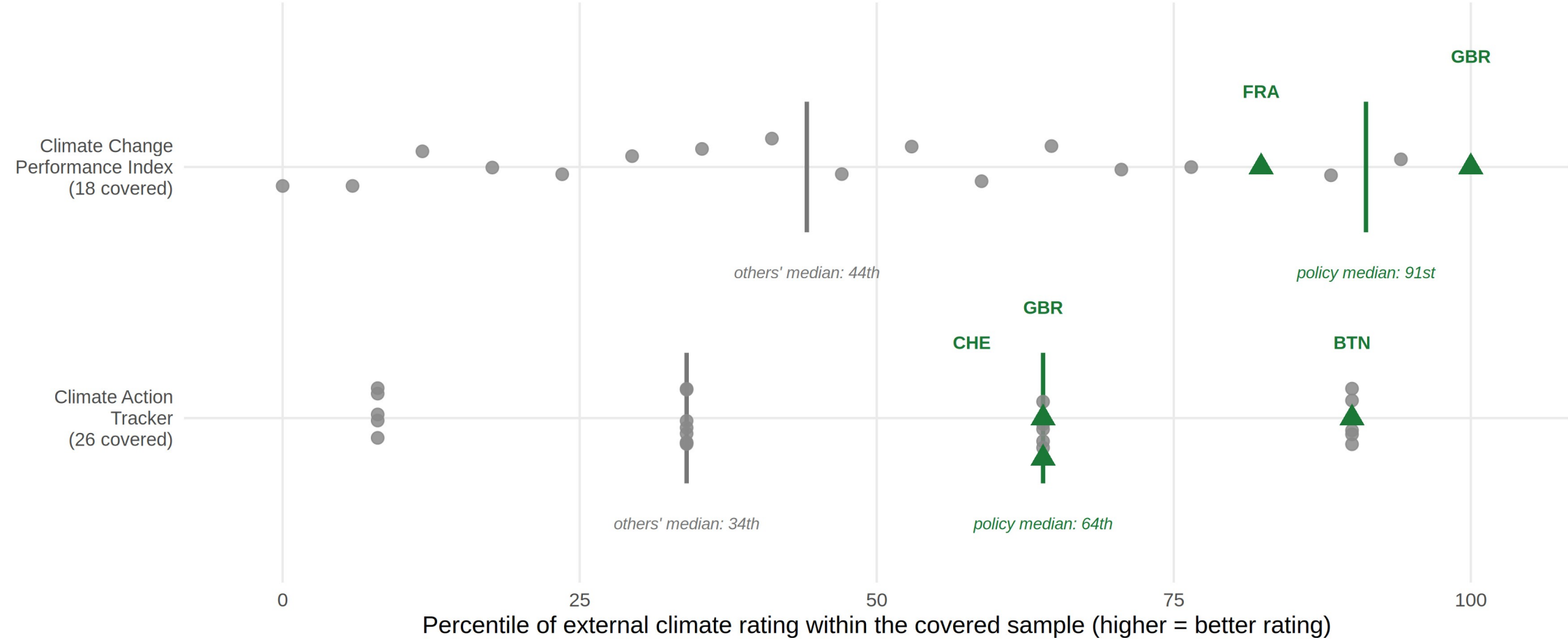
Mechanisms are regional stories; success concentrates where policy changed physical systems



Independent raters recognize the same countries — on thin coverage

Independent climate raters rank the policy-enabled cases high, but coverage is thin

External-rating percentile among covered countries. Green triangles: policy-enabled cases (CAT covers 3 of 9; CCPI covers 2). Corroborative, not definitive.



Robustness, corroboration, and honest limits

Consumption-based accounting

Of 17 countries eligible under both accountings, only 6 agree within 5 years; the median gap is ~12 years. Some 'domestic' transitions partly reflect trade and offshoring — a main result, not a footnote. (Consumption series begin in 1990, so very early production breaks cannot be matched by construction.)

Carbon pricing & policy stringency

Pricing is negatively associated with C/E in country-and-year fixed-effects models (contemporaneous strongest); C/GDP and E/GDP estimates are imprecise. OECD EPS covers only 15 of 28 focal countries. Associational, not causal.

Event-timing validation

45% of documented events within 2 years of the break vs ~17% by chance; 80% within 5 vs 37%; 70% inside the episode vs 33% — far beyond retrospective storytelling.

What breakpoints can't tell you

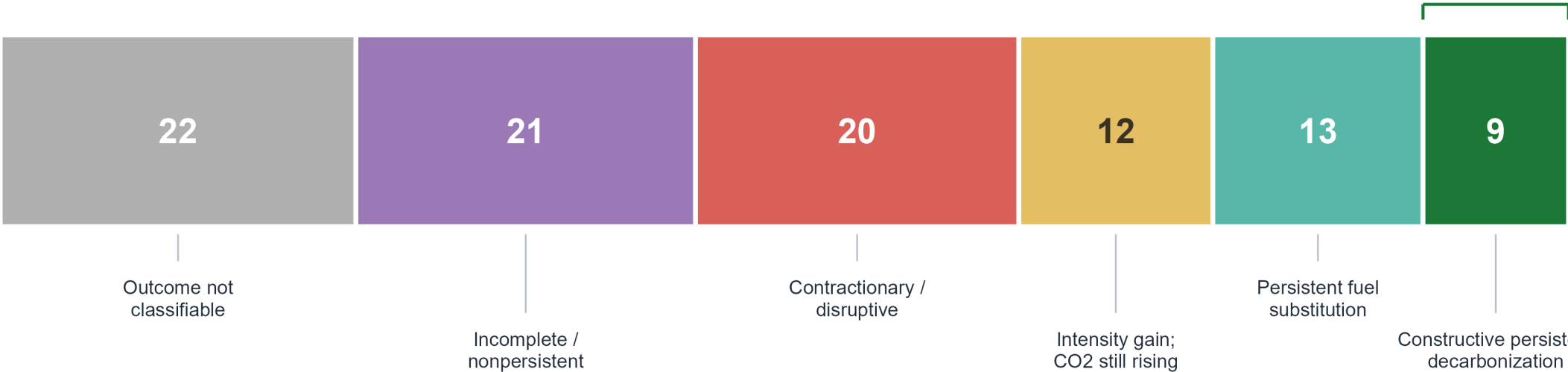
Break timing identifies change, not cause. Attribution and outcome classification each leave 22 cases open — restraint that makes the resolved 75% credible. No counterfactual: a treaty that prevents backsliding produces no break at all.

THE VERDICT

Curves bend everywhere; decarbonization is rare

97 statistically eligible national breaks —
every one a real, favorable-looking bend

only 9 pair a persistent intensity decline
with falling absolute CO2



Global agreements can organize climate action — durable national transformations bend the curve.

- Global curves bent in the 1970s; since ~2013 they bend faster — a real supply-side acceleration whose timing cannot yet separate Paris from the renewables revolution it helped consolidate.
- Treaty-adjacent signals exist — especially around Rio — but few survive placebo scrutiny, and national transitions scatter across five decades.
- Favorable bends often reflect recession, disruption, deindustrialization, or offshoring; intensity gains are routinely overwhelmed by growth.
- Nine countries pair persistent intensity declines with falling absolute CO2 — policy made physical in fuel mix, infrastructure, and economic structure.

*Structural breaks say when · Kaya says through which pathway · historical validation says what plausibly happened ·
transition quality says whether it mattered*